

Luiz Mugnaini

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EXPERIENCE

Founder & Lead Developer

Presheaf Studio

April 2024 - Present

São Paulo, Brazil

- Designing a native cross-platform application for professional digital sculpting based on my research on polygonal mesh retopology.
- Creating novel SIMD-optimized algorithms with multi-threading capabilities for mesh editing.

Software Development Engineer (SDE)

Amazon

February 2025 - Present

São Paulo, Brazil

- Spearheaded an invoice processing system for the Indian government handling 1 million updates per day.
- Led the implementation of the new transitive token security protocol for spec-compliance of our global invoicing services.
- Ported the legacy testing framework of Amazon's world-wide store-front website for all of our invoice solutions.
- Worked on-call for our invoice pipeline with direct contact with both clients and third-party suppliers.

Fullstack Python Contractor

NDA startup

June 2023 - March 2024

São Paulo, Brazil

- Built a full web app for medical ultrasound courses hosted on Google Cloud.
- Used FastAPI and HTMX with Jinja2 templates and Tailwind to build a fluid user experience.
- Implemented user authentication using SQLite and a distributed asset versioning system using S3.

EDUCATION

University of São Paulo

Master's in Computer Graphics (GPA -/-)

São Paulo, Brazil

Aug. 2024 - Present

I research novel techniques for geometry processing of 3D meshes leveraging ideas from the growing field of Discrete Differential Geometry. As part of my work, I'm developing algorithms for better re-topology workflows.

University of São Paulo

BSc in Molecular Sciences (GPA 9.2/10)

São Paulo, Brazil

Jul. 2020 - Aug. 2024

Interdisciplinary scientific research in Computer Science and Mathematics, originating in combinatorial models of Homotopy Theory, my studies evolved into the discretization of continuous surfaces and its applications to Computer Graphics. This led me into researching differential methods for the geometry processing of polygonal meshes. I've documented all of my studies in a comprehensive compendium of Mathematics and Computer Science.

Relevant course-work: Algorithms, Data Structures, Computer Graphics, Differential Geometry, Algebraic Topology.

PROJECTS

runtime | C++, C, Windows, Linux

Oct. 2023 - Present

C11/C++20 library used as a high-performance alternative to the STL, containing the implementation of custom allocators, core algorithms, foundational data structures, OS compatibility layer, and a hardware abstraction layer. This is the predecessor of the current Presheaf Studio in-house runtime library.

Yet unnamed game | C++, C, Vulkan, GLSL, Lua, Windows

Oct. 2023 - Present

2D pixel-art game written from scratch in C++ with a Vulkan-based rendering backend, and minimal dependencies. Implemented a multi-threaded hot-reloading system for the gameplay code for faster iteration times.

mina | C++, C, GLFW, Vulkan, GLSL, Lua

Apr. 2024 - Paused

Cross-platform Game Boy emulator in C++ with WinAPI for Windows and GLFW for Unix-based systems. Explored the interaction of multiple device components with integrated CPU, graphics and audio pipelines.

chirp | Rust, SDL, Emulation, Interpreters

Dec. 2022 - Dec. 2022

Emulator for the CHIP-8 virtual-machine, written in Rust with SDL. Researched techniques for emulating hardware and building interpreters.

radiant | Rust, SDL

Nov. 2021 - Dec. 2021

Multi-threaded CPU-based ray tracer with SDL as the rendering backend.

TECHNICAL SKILLS

Programming languages: C++, C, Rust, Python, Lua, Java, Typescript, Javascript.

Libraries: Modern C++ STL, C11 stdlib, WinAPI, Linux ABI, Vulkan, OpenGL, SDL.